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DEPARTMENT OF

INSTRUMENTATION AND CONTROL ENGINEERING
(2024 – 2025)

ARDUINO BASED VERTICALLY WALL CLIMBING GLASS CLEANING ROBOT

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PROJECT WORK

(2024 – 2025)



BONAFIDE CERTIFICATE

It is certified that the project entitled **"ARDUINO BASED VERTICALLY WALL CLIMBING GLASS CLEANING ROBOT"** is the bonafide work done by _____ in partial fulfillment of the requirements for the award of **"Diploma in Instrumentation and Control Engineering"** during the year 2024-2025.

Submitted for the board practical examination held on _____

PROJECT GUIDE **HEAD OF THE DEPARTMENT**

INTERNAL EXAMINER **EXTERNAL EXAMINER**

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1. ABSTRACT

Nowadays, modern houses and offices have been constructed with a large number of curtain window glass walls. Human faces a lot of difficulties for its maintenance, repair, and care from dust and pollution because of its height and the exposure to the risk of hurt or injury during the cleaning process. Many kinds of cleaning mechanisms are invented to ease the human daily chores, such as vacuum cleaners and window cleaners, which are used to clean glass windows. Nowadays, with the large increase in the development of tall and smart buildings in urban areas, the window-cleaning robot becomes a necessity. Robots have been created to assist or replace humans in various dangerous and difficult tasks. The window-cleaning robot is one of the robots that have emerged in recent decades. This robot can be used in homes, offices, and large buildings. The main objective of this project is to design a robot that can clean glass windows efficiently and rapidly even in dangerous and hazardous places. The robot will be controlled using the ARDUINO microcontroller. The ultrasonic sensor senses the ultrasonic waves reflected from the surface and gives an input signal to the ARDUINO board, which will identify the danger for the robot movement. A wiper is attached to the servo motor to clean the window glass. A BLDC motor fan is used to create a vacuum to stick to the glass window. A hobby gearmotor is used for wheel movement of the robot. LEDs and a buzzer are used to indicate the robot's danger occurred for it. LiPo batteries are used for the movement and working of robot.

2. INTRODUCTION

The maintenance of high-rise buildings presents numerous challenges, particularly in the critical task of window cleaning. Traditional methods often involve significant risks for workers, logistical complexities and can be labor-intensive.

This project addresses the need for a safer, more efficient, and potentially autonomous solution by presenting the design and implementation of a wall-climbing window-cleaning robot. This innovative approach aims to mitigate the challenges associated with conventional techniques by introducing a robot capable of adhering to vertical surfaces, navigating across them, and performing cleaning tasks.

The development of this robot explores the integration of mechanical design, electronic control, and software programming to create a functional prototype. Utilizing an ARDUINO microcontroller, various sensors, and a custom-designed climbing and cleaning mechanism, the robot demonstrates the potential for autonomous operation in challenging environments.

This report provides a comprehensive overview of the robot's development process, including problem definition, hardware selection, software algorithms, and testing procedures, highlighting its potential to revolutionize building maintenance Practices and improve worker safety.

2.1 Existing Solutions

The challenge of cleaning windows in high-rise buildings has led to the development of various solutions, each with its own advantages and limitations. Traditional methods often involve manual labor, utilizing scaffolding, suspended platforms, or workers rappelling down the building. While these methods are still widely used, they present significant safety risks for workers, logistical complexities and can be time-consuming and labor-intensive.



Fig 1. Window cleaning by humans

In response to these challenges, there has been increasing interest in automated solutions, particularly wall-climbing robots. These robots offer the potential to improve safety, increase efficiency, and reduce labor costs. Existing wall-climbing robots employ various technologies to achieve adhesion and locomotion, including

- **Suction Mechanisms:** Some robots utilize suction cups or vacuum systems to adhere to smooth surfaces like glass.
- **Magnetic Wheels:** For ferromagnetic surfaces, magnetic wheels can provide a strong adhesive force.

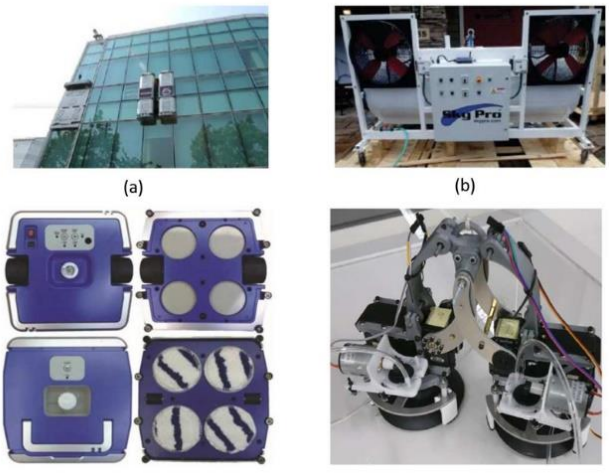


Fig 2. Different adhesion methods of robot

2.2 Technologies Used

The development of a wall-climbing window-cleaning robot involves the integration of several key technologies:

- **Locomotion Methods:** Wall-climbing robots employ various methods for movement, such as wheeled motion or tracked systems. The design of the locomotion system is crucial for maneuverability and stability on vertical surfaces.

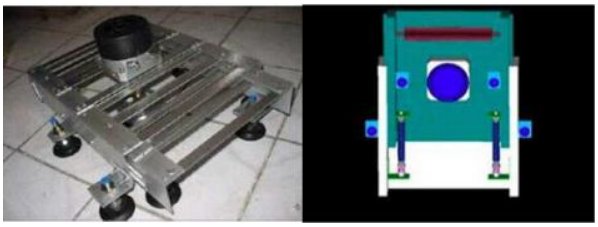


Fig 3. A cleaning robot based on parallel transmitional mechanism

- **Control Systems:** Microcontrollers, such as the ARDUINO, are often used to control the robot's movement, cleaning mechanisms, and sensor systems. These control systems process sensor data and generate commands to actuate the robot's various components.

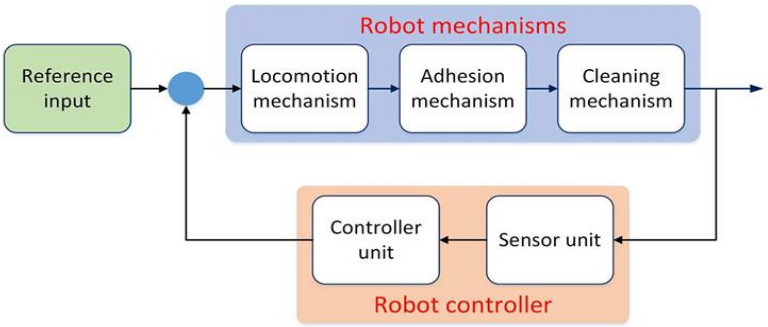


Fig 4. Diagram of control system for window cleaning robot

- **Sensing:** To navigate and operate autonomously, wall-climbing robots may utilize various sensors, including proximity sensors, gyroscopes, and cameras. These sensors provide information about the robot's environment and orientation.

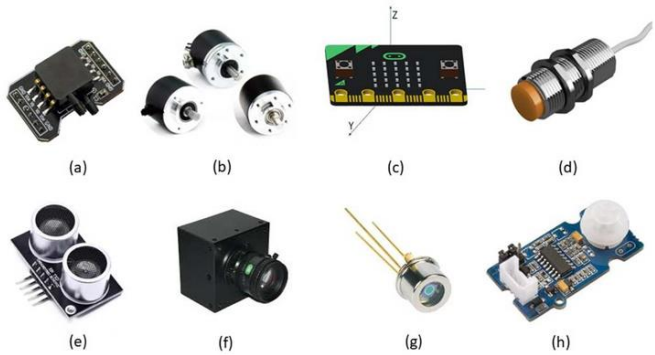


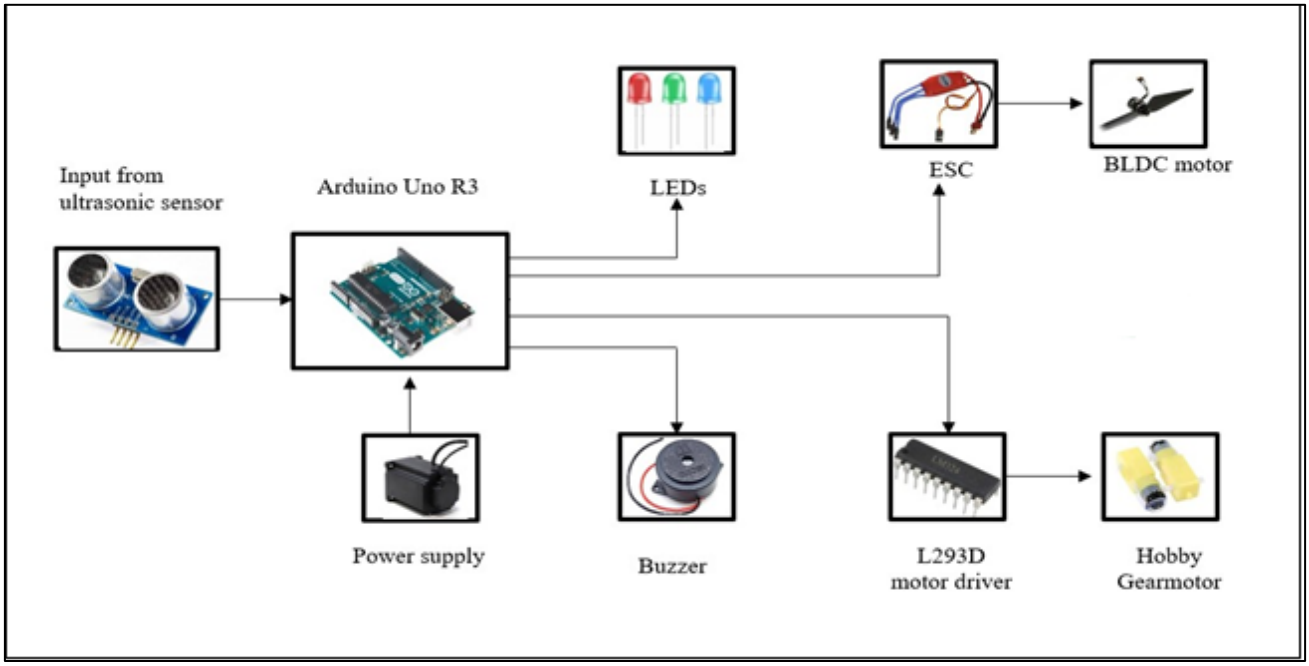
Fig 5. Various essential sensors used in window cleaning robot.

- **Cleaning Mechanisms:** Different cleaning mechanisms can be integrated into wall-climbing robots, such as rotating brushes, spraying systems, or wiping devices. The choice of cleaning mechanism depends on the specific cleaning requirements.



Fig 6. Cleaning mechanism based on two degree of freedom parallel manipulator.

3. BLOCK DIAGRAM



3.1 COMPONENTS

1. ARDUINO UNO R3
2. Ultra sonic sensor (HC-SR04)
3. BLDC motor (2450 kv*)
4. N20 Hobby gearmotor (60 RPM)
5. Buzzer and LED
6. L298n Motor Driver
7. Jumper Wires and Connectors.
8. 12v LiPo Battery

The wall-climbing robot operates as an integrated system, orchestrated by the ARDUINO Uno R3 microcontroller. The system's functionality can be broadly divided into sensing, processing, and actuation.

**In a BLDC motor, "kV" (not to be confused with kilovolts) stands for RPM per Volt — it tells you how fast the motor will spin per volt of input with no load.*

Sensing: The robot utilizes an ultrasonic sensor to perceive its environment. The ultrasonic sensor emits high-frequency sound waves and measures the time taken for the echoes to return after bouncing off an obstacle. This time measurement is then used to calculate the distance to the obstacle, providing crucial input for obstacle avoidance.



Processing: The ARDUINO Uno R3 serves as the central processing unit. It receives the distance data from the ultrasonic sensor, as well as control signals from the user via Bluetooth communication and a potentiometer. The Arduino executes a pre-programmed code that interprets the sensor data, makes decisions based on the defined algorithms, and generates control signals for the various actuators.



Actuation: The ARDUINO controls several key actuators to enable the robot's movement, wall adhesion, and cleaning functions.

- **BLDC Motor and ESC:** A Brushless DC (BLDC) motor, in conjunction with an Electronic Speed Controller (ESC), generates the suction force necessary for the robot to adhere to the vertical surface. The Arduino sends a Pulse Width Modulation (PWM) signal to the ESC, which regulates the power delivered to the BLDC motor, thereby controlling the suction intensity.

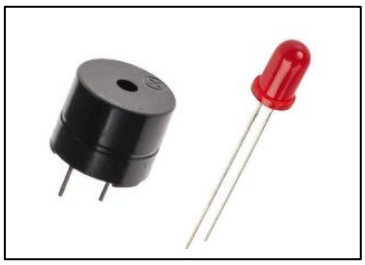


- The user can adjust the BLDC motor speed using a through app connected to ARDUINO.

- **N20 Motors and L298n Motor Driver:** The robot employs hobby gearmotors (likely N20 motors) for locomotion, enabling it to move across the surface. An L298N motor driver is used to control the direction and speed of these motors. The Arduino sends control signals to the L298N, which amplifies the signals to drive the motors.



- **Buzzer and LEDs:** A buzzer is included to provide audio feedback, such as alerts or status indicators. LEDs are also used for visual feedback, providing information about the robot's operation. The ARDUINO controls these output devices.

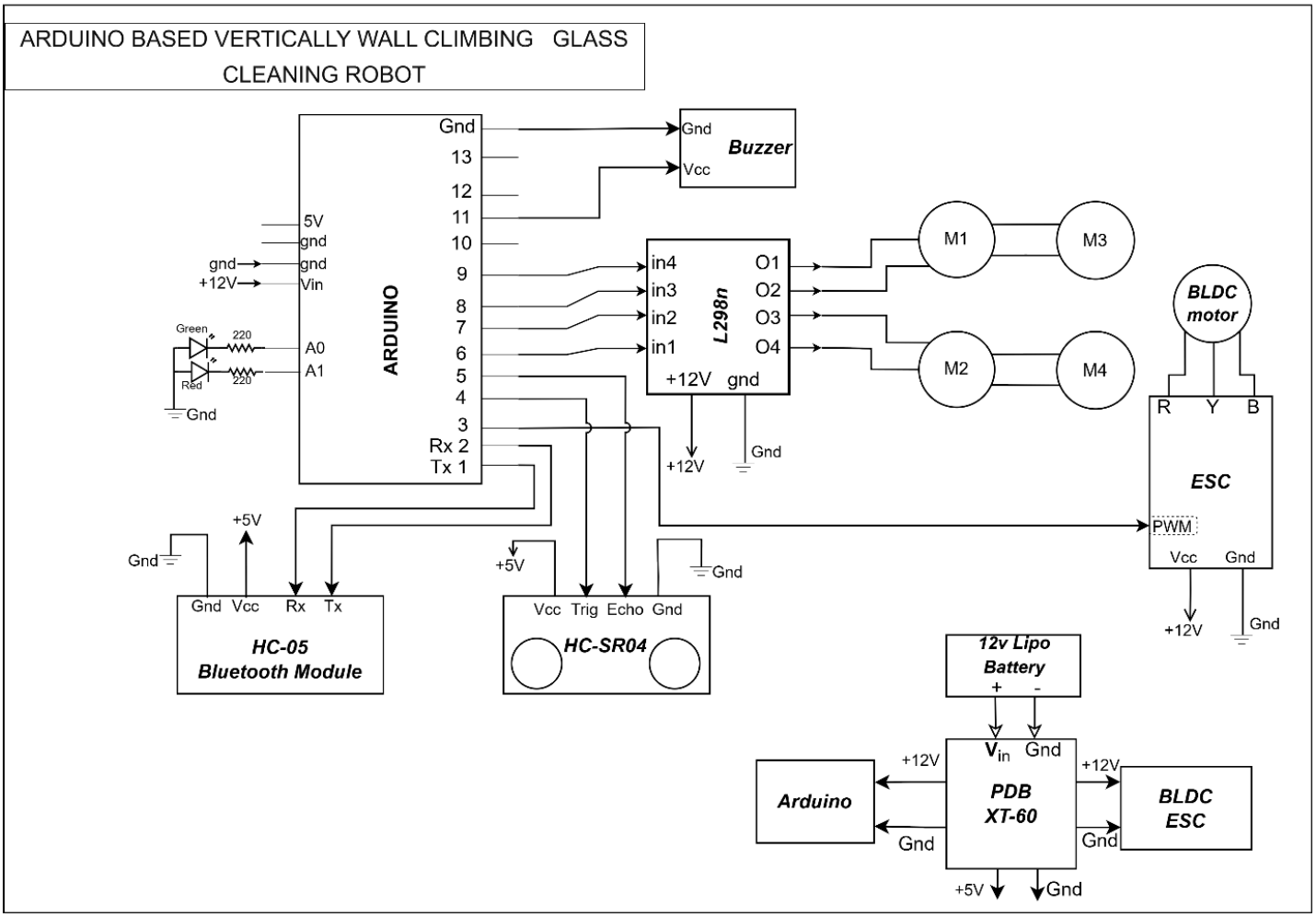


Power Supply: The entire system is powered by a LiPo battery, which provides the necessary electrical energy for all the components to operate. This ensures a stable and reliable power source for the robot's functionality.

In summary, the robot's operation involves continuous sensing of the environment, processing of the sensed information by the ARDUINO, and actuation of the various components to achieve controlled movement, wall adhesion, and cleaning.



4. Circuit Diagram



Circuit Overview

The circuit of the wall-climbing glass-cleaning robot is divided into multiple functional sections, each responsible for a specific operation. The central control unit is the ARDUINO Uno R3, which coordinates sensing, actuation, communication, and power management.

1. Power Supply Section

The robot is powered by a 12V 3S 30C LiPo battery connected via an XT60 connector to a Power Distribution Board (PDB). The PDB efficiently distributes power to all major components as follows:

- **ARDUINO Uno R3** receives 12V to its VIN pin.

- **The L298N motor driver** is powered by a separate 12V line to drive the four N20 DC motors.
- **The Simonk 30A ESC** receives 12V directly from the PDB to control the BLDC motor.
- **The HC-SR04 ultrasonic sensor** is powered using the regulated 5V output from the ARDUINO.

This configuration ensures isolation of logic and power circuits and minimizes voltage drops and power noise.

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2. Central Control Unit—ARDUINO Uno R3

The ARDUINO Uno R3 serves as the brain of the robot. It receives input from sensors, processes the data, and generates control signals to drive actuators. The ARDUINO is responsible for:

- Controlling the **L298N motor driver** to operate the N20 DC motors.
- Receiving obstacle detection signals from the **HC-SR04 ultrasonic sensor**.
- Sending PWM signals to control the **buzzer** and **LED indicators**.
- Managing Bluetooth communication with the **HC-05 module**.
- Optionally controlling a **servo motor** to actuate the cleaning mechanism.

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3. Motor Driver Section—L298N

The L298N motor driver is used to control the four N20 gear motors arranged in pairs (left and right sides). The ARDUINO provides digital control signals (IN1–IN4) to determine motor direction and PWM signals (ENA and ENB) to control speed. The outputs of the L298N are connected to the motors in the following way:

- **OUT1/OUT2**: Left-side motor pair
- **OUT3/OUT4**: Right-side motor pair

This setup allows the robot to move forward, backward, and turn using differential drive.

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4. Suction System—BLDC Motor with ESC

A 2450 KV BLDC motor is used to generate downward airflow for suction. It is connected to a 1045 propeller and is controlled via a 30A Simonk ESC. The ESC receives:

- Power directly from the 12V LiPo battery via the PDB
- A PWM control signal from either the ARDUINO or a servo tester

This motor enables the robot to stick to vertical glass surfaces using suction force.

---

5. Obstacle Detection and Indication

An **HC-SR04 ultrasonic sensor** is mounted on the front of the robot for obstacle detection. It sends trigger pulses and measures echo time to determine the distance from obstacles. The ARDUINO processes this data, and if an obstacle is within a threshold range, it triggers

- A **buzzer** (audible alert)
- **Red and green LEDs** (visual status indicators)

This system helps avoid collisions or unsafe conditions during operation.

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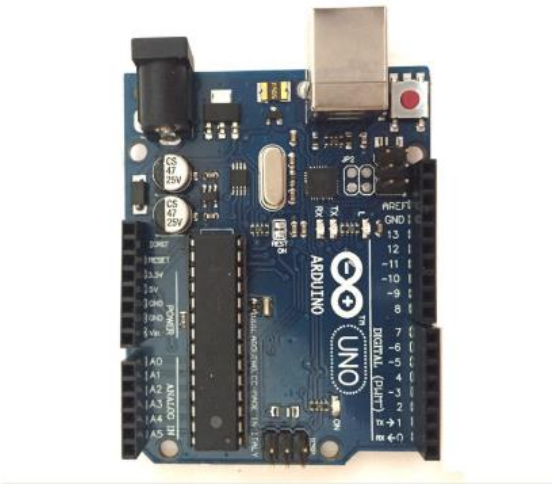
6. Bluetooth Communication

An **HC-05 Bluetooth module** is used to receive wireless control commands from a smartphone app. It is connected to the ARDUINO's serial communication pins (TX and RX). Commands such as forward, backward, left, right, and stop are sent via Bluetooth and processed by the ARDUINO to control the robot's motion accordingly.

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5. Hardware

5.1 ARDUINO Uno R3



The ARDUINO Uno R3 is a versatile open-source microcontroller board.

Feature	Specification	Details
Microcontroller	Chip	ATmega328P
	Architecture	AVR
	Flash Memory	32 KB
	SRAM	2 KB
	EEPROM	1 KB
	Clock Speed	16 MHz
Input/Output	Digital I/O Pins	14
	Analog Input Pins	6
	PWM Output Pins	6
	Serial (UART) Ports	1
	SPI	Yes
	I2C	Yes
Power	Recommended Input Voltage	7-12V
	Input Voltage Limits	6-20V
	5V Output Current	Up to 500 mA
	3.3V Output Current	50 mA
	DC Current per I/O Pin	20 mA
	DC Current for 3.3V Pin	50 mA
Other	USB Connection	USB Type B
	ICSP Header	Yes
	Reset Button	Yes
	Dimensions	2.7 inches x 2.1 inches

Applications and Use in our Project:

The ARDUINO Uno R3 is a popular choice for a wide range of applications due to its ease of use, versatility, and affordability. Here are some common applications:

- **Robotics:** Controlling motors, reading sensors, and making decisions for robots.
- **Home Automation:** Controlling lights, appliances, and security systems.
- **Data Logging:** Recording sensor data for analysis.
- **Prototyping:** Quickly creating and testing electronic circuits and systems.
- **Educational Purposes:** Teaching and learning about electronics and programming.

In the project, the ARDUINO Uno R3 serves as the "brain" of the robot.

We are using it to:

- **Control Motors:** The ARDUINO sends signals to the L298N motor driver to control the N20 motors that move the robot on the wall.
- **Control Steering:** The ARDUINO controls the servo motor to steer the robot.
- **Read Sensors:** The ARDUINO reads data from the HC-SR04 ultrasonic sensor to detect obstacles.
- **Communicate via Bluetooth:** The ARDUINO communicates with a smartphone via the HC-05 Bluetooth module, allowing for remote control of the robot.

Essentially, the ARDUINO Uno R3 processes information from sensors, makes decisions based on its programming, and sends signals to actuators (motors, etc.) to control the robot's movement and cleaning functions.

5.2 Ultrasonic Sensor (HC-SR04) Details



HC-SR04 Ultrasonic Sensor Pinout

The HC-SR04 is a widely used, low-cost ultrasonic sensor module that enables non-contact distance measurement. It's a popular choice for various projects, including robotics, automation, and interactive installations.

Specifications

Feature	**Specification**	**Details**
Operating Voltage	+5V DC	
Current Consumption	15 mA (typical)	
Frequency	40	
Measuring Angle	15 degrees	
Ranging Distance	2cm - 400cm	
Resolution	3mm	
Accuracy	±1cm	
Pin Details	VCC	+5V Power Supply
	GND	Ground
	Trig	Trigger Input
	Echo	Echo Output

Distance Calculation

The HC-SR04 calculates distance by measuring the time it takes for an ultrasonic sound wave to travel to an object and return. Here's a more detailed explanation:

1. **Triggering:** The ARDUINO sends a short pulse (typically 10 microseconds) to the HC-SR04's Trig pin.
2. **Transmission:** The sensor emits an 8-cycle burst of ultrasonic sound waves at 40 kHz.
3. **Reception:** The sensor's receiver listens for the echo of these waves.
4. **Echo Pulse:** The echo pin goes HIGH when the sound is transmitted and stays HIGH until the echo is received. The duration the echo pin is HIGH represents the sound's travel time.
5. **Time Measurement:** The ARDUINO measures this "HIGH" time from the Echo pin.
6. **Calculation:** The distance is calculated using

Distance = (Time x Speed of Sound) / 2

- **Time:** The time the Echo pin is HIGH (in seconds).
- **Speed of Sound:** Approximately 343 meters per second (m/s) in dry air at 20°C. This can vary slightly with temperature and humidity.
- **Division by 2:** The sound wave travels to the object AND back, so we divide by 2 to get the one-way distance.

Applications

- Obstacle avoidance in robotics
- Distance measurement
- Level sensing
- Proximity detection
- Object counting

Use in Our Project

In our wall-climbing window-cleaning robot, the HC-SR04 is used for obstacle detection:

- The sensor is mounted on the robot.
- The ARDUINO triggers the sensor and reads the Echo pin.
- The ARDUINO calculates the distance and compares it to a threshold (5cm).
- If an obstacle is detected within the threshold:
    - The ARDUINO stops the robot's movement.
    - The ARDUINO activates a buzzer.

5.3 Passive Buzzer



Introduction

A passive 5V buzzer is an electromechanical transducer employed to generate auditory signals by converting an externally supplied alternating current (AC) signal into sound. The frequency of this signal determines the acoustic pitch. Its selection is based on its fundamental design and capacity for tonal variation through external control.

Key Specifications

Characteristic	Typical Value(s)	Specification Notes
Electrical Properties		
Rated Voltage	5V DC	Nominal operating voltage.
Operating Voltage Range	3V to 6V DC	Refer to datasheet for precise limits.
Current Consumption	< 30 mA	At rated voltage; varies with frequency.
Resonant Frequency	2000 Hz to 3000 Hz	Frequency for optimal sound output; consult datasheet.
Acoustic Properties		
Sound Pressure Level	≥ 80 dB at 10 cm	Measured at specified conditions.
Tone Generation	Variable (external signal dependent)	Output depends on the external AC signal.
Mechanical Properties		
Transducer Type	Passive	Requires external oscillating signal.

Physical Dimensions	Small (typically 8-12mm diameter)	Refer to datasheet for precise dimensions.
Operational Requirement		
Excitation Signal	External AC Signal Required	DC voltage alone will not produce sound.

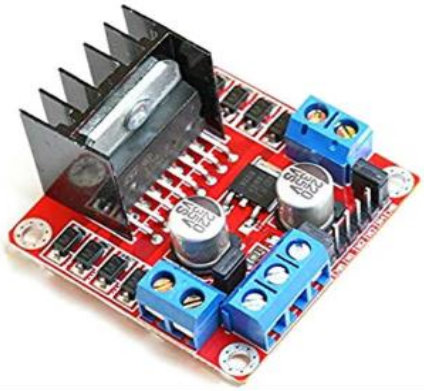
Applications of Passive Buzzers:

- Alarm Systems,
- User Interface Feedback,
- Audio Indicators,
- Educational Demonstrations,
- Notification Devices.

Use in Our Project

In this project, a passive 5V buzzer provides auditory feedback upon obstacle detection by the ultrasonic sensor. The microcontroller generates an appropriate oscillating signal to the buzzer based 3on the sensor's readings.

5.4 L298N Motor Driver Details



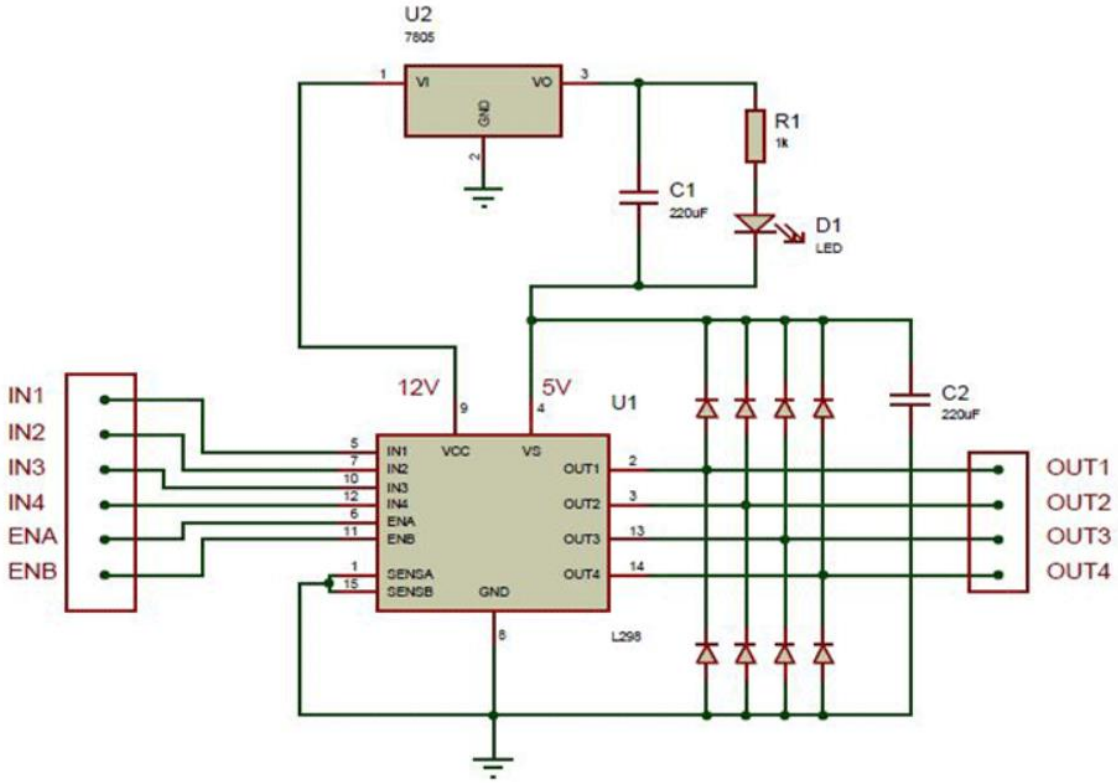
The L298N is a popular motor driver integrated circuit (IC) that allows microcontrollers like the ARDUINO to control the speed and direction of DC motors. It's commonly used in robotics projects.

Specifications

Feature	Specification	Details
Supply Voltage (Vs)	4.8V - 35V	
Motor Supply Current	Up to 2A per channel (peak)	With adequate heat sinking. Continuous current is lower.
Logic Supply Voltage (Vss)	5V - 7V	Used to power the L298N's logic circuitry. Often, the ARDUINO's 5V supply is used.
Logic Current	0-36mA	Current drawn by the L298N's logic.
Driver Output	Dual H-bridge	Can drive two DC motors or one stepper motor.
Max.Power Dissipation	25W	Requires heat sinking at higher currents.
Control Signal Voltage	5V	
Pin Details	VCC	Logic supply voltage (+5V)
	GND	Ground
	Vs. (VMotor)	Motor supply voltage (+4.8V to +35V)

	ENA, ENB	Enable inputs for motor channels A and B (PWM for speed control)
	IN1, IN2	Inputs for controlling the direction of Motor A
	OUT1, OUT2	Outputs for Motor A
	IN3, IN4	Inputs for controlling the direction of Motor B
	OUT3, OUT4	Outputs for Motor B
	SenseA, SenseB	Current sensing (Connect to GND through a small resistor)

Internal circuit diagram of L298N Motor Driver module is given below:



Working Principle

The L298N contains two H-bridge circuits. Each H-bridge can control one DC motor.

1. **H-Bridge Operation:**
   - An H-bridge consists of four switches (typically transistors).
   - By controlling these switches, the direction of current flow through the motor can be reversed.
   - For example:
     - To make a motor spin forward, switches 1 and 4 are turned ON.

        - To make it spin backward, switches 2 and 3 are turned ON.
    - Turning on switches 1 & 3 or 2 & 4 would short-circuit the power supply and must be avoided.
2. **Direction Control:**
    - The L298N uses digital inputs (IN1, IN2, IN3, IN4) to control the switches in the H-bridges.
    - By setting the appropriate combination of HIGH and LOW signals on these input pins, you can control the motor's direction (forward, backward, or stop).
3. **Speed Control:**
    - The L298N has Enable (ENA, ENB) pins.
    - By applying a Pulse Width Modulation (PWM) signal to these pins, can vary the average voltage applied to the motors, thus controlling their speed.
    - A higher duty cycle means a higher average voltage and faster speed.
4. **Current Sensing:**
    - The L298N has SenseA and SenseB pins.
    - By connecting a small resistor between these pins and ground, can measure the voltage across the resistor, which is proportional to the motor current.
    - This allows for current limiting or monitoring.

Applications

- **Robotics**: Controlling the movement of robots.
- **RC Vehicles**: Driving motors in cars, trucks, and other vehicles.
- **Automation**: Controlling motors in industrial equipment.
- **CNC Machines**: Driving stepper and DC motors for precise control.

Use in Our Project

In our wall-climbing window-cleaning robot, the L298N motor driver is used to control the movement of the robot.

- The ARDUINO Uno sends digital signals to the IN1, IN2, IN3, and IN4 pins of the L298N to control the direction of the two pair of N20 DC motors.
- The ARDUINO sends PWM signals to the ENA and ENB pins of the L298N to control the speed of the N20 DC motors.
- This allows the robot to move forward, backward, rotate right left and stop, enabling it to traverse the window surface.

5.5 A2212 2450 KV BLDC Brushless DC Motor



2212 Brushless Motor



2212 Brushless Motor Pinout

A 2450KV BLDC motor is a type of brushless DC electric motor that is commonly used in applications requiring high speed and efficiency, such as drones, RC aircraft, and other robotic systems.

Specifications

Feature	Specification	Details
KV Rating	2450 KV	This is the motor's speed constant. It indicates the RPM (revolutions per minute) per volt applied.
Operating Voltage	7.4V - 11.1V (2S-3S LiPo)	Typical voltage range.
Max Current	Varies (e.g., 20-30A)	This depends on the specific motor and load.
Max Power	Varies (e.g.,200-300W)	Depends on voltage and current limits.
Stator Diameter	Varies (e.g., 22mm)	Dimension of the motor's core.
Stator Length	Varies (e.g., 12mm)	Dimension of the motor's core.
Wiring	3 Wires	Connect to ESC (Electronic Speed Controller). The order may affect rotation direction.

*****_In a BLDC motor, "kV" (not to be confused with kilovolts) stands for RPM per Volt — it tells you how fast the motor will spin per volt of input with no load._

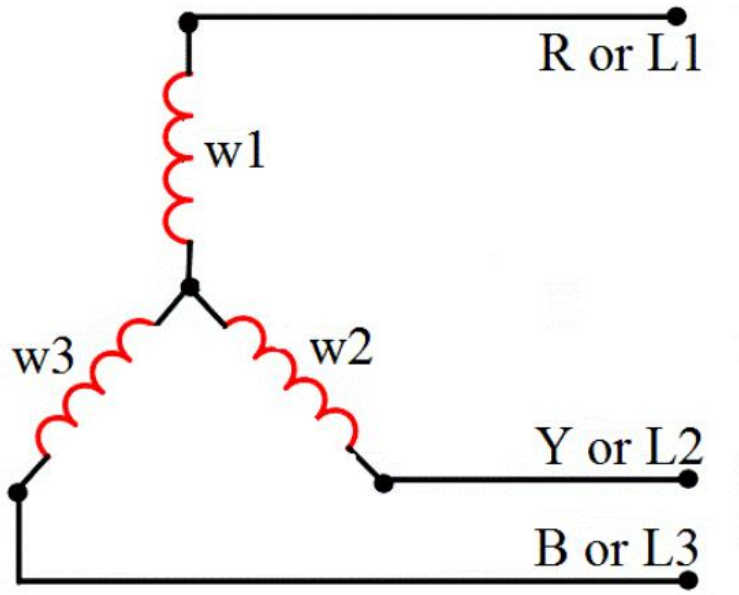
Understanding the KV Rating

The KV rating is crucial:

- A 2450KV motor will, in theory, rotate at 2450 RPM for every volt applied.
- For example, with an 11.1V (3S LiPo) battery, the no-load RPM would be approximately 2450 KV * 11.1V = 27,200 RPM.
- In practice, the actual RPM under load will be lower due to factors like friction and propeller load.
- A higher KV rating means the motor spins faster per volt, but it may produce less torque.

Brushless Motor Pin Configuration

The motor has three pins as shown in figure. The internal connection between three of them is shown below.



Pin Name	Description
R or L1	Connected to winding1 or phase1
Y or L2	Connected to winding2 or phase2
B or L3	Connected to winding3 or phase3

*Remember there is no neural terminal driven out from the winding connections.

Working Principle

BLDC motors operate differently than brushed DC motors:

1. **Electronic Commutation:** Instead of brushes, BLDC motors use electronic commutation. The ESC (Electronic Speed Controller) energizes the motor's stator windings in a specific sequence.
2. **Stator and Rotor:**
    - The stator is the stationary part of the motor and contains the windings.
    - The rotor is the rotating part and contains permanent magnets.
3. **Magnetic Interaction:** The ESC creates a rotating magnetic field in the stator. This field interacts with the magnetic field of the rotor's permanent magnets, causing the rotor to rotate.
4. **ESC Control:** The ESC precisely controls the timing and sequence of the stator winding energization, which controls the motor's speed and direction.
5. **Feedback (Sometimes):** Some BLDC systems use sensors (like Hall sensors) to provide feedback to the ESC about the rotor's position. This allows for more precise control. They rely on "sensorless" ESCs.

Advantages of BLDC Motors

- High efficiency
- High power-to-weight ratio
- Long lifespan (no brushes to wear out)
- High speed capability

Applications

- Drones and multirotors
- RC aircraft
- Electric vehicles
- Robotics
- Industrial automation

Use in Our Project

In our wall-climbing window-cleaning robot, the 2450KV BLDC motor is used to generate suction.

- The BLDC motor is connected to a propeller.
- The ESC controls the speed of the BLDC motor.
- As the propeller spins, it creates a downward airflow, generating suction.
- This suction force is used to attach the robot to the window surface, allowing it to climb.

5.6 Simonk 30A ESC



The Simonk 30A ESC (Electronic Speed Controller) is a device that controls the speed of a brushless DC (BLDC) motor. It is commonly used in multirotors (drones) and other RC applications where precise and efficient motor control is required. Simonk refers to a specific firmware designed for improved performance in these applications.

Introduction

An ESC is a crucial component in any BLDC motor system. Unlike a DC motor, a BLDC motor cannot be directly controlled by a simple voltage. It requires a specific sequence of electrical signals to rotate, and the ESC provides these signals. The simonk firmware is an open-source firmware designed to provide very fast response, which is especially important for multirotors.

Specifications

Feature	Specification	Details
Continuous Current	30A	The maximum current the ESC can handle continuously.
Burst Current	Varies (e.g., 40A for 10 seconds)	The maximum current the ESC can handle for a short period.
Input Voltage	7.4V - 11.1V (2S-3S LiPo)	Typical voltage range.
BEC Output	None	This ESC does *not* have a Battery Eliminator Circuit to provide power to the receiver/microcontroller. You'll need a separate 5V power source.
PWM Input Signal	50-500 Hz	Standard PWM signal range from a receiver or microcontroller.
Firmware	Simonk	Optimized for fast response in multirotor applications.
Size	Varies	Compact size; dimensions vary by manufacturer.
Weight	Varies	Lightweight design.
Wiring	Battery Input	Thick-gauge wires, typically with a connector (e.g., XT60).
	Motor Output	Three wires are connected to the BLDC motor.
	Signal Input	Three-wire connector (signal, +, -), connected to the receiver or microcontroller.

Working Principle

The Simonk 30A ESC takes a control signal from a receiver or microcontroller and uses it to regulate the speed of a BLDC motor. Here's how it works:

1. **Input Signal:** The ESC receives a PWM (Pulse Width Modulation) signal.
    - A standard PWM signal varies the width of a pulse.
    - The pulse width typically ranges from 1000 microseconds (µs) to 2000 µs.
    - 1000 µs usually represents the minimum speed (or stopped), and 2000 µs represents the maximum speed.

2. **Electronic Commutation:** The ESC uses the input signal to control the switching of its internal transistors (usually MOSFETs).
    - It generates a three-phase AC signal that drives the BLDC motor.
    - The ESC switches the power to the motor windings in a specific sequence, creating a rotating magnetic field.
    - This rotating field interacts with the BLDC motor's magnets, causing the motor to rotate.
3. **Speed Control:** The ESC varies the frequency and duty cycle of the three-phase signals sent to the motor, effectively controlling its speed.
    - A higher input signal (longer pulse width) tells the ESC to switch the transistors faster, resulting in a higher motor speed.
4. **Simonk Firmware:** The Simonk firmware is designed for:
    - **Fast Response:** It reacts quickly to changes in the input signal, which is crucial for stabilizing multirotors.
    - **High Efficiency:** It minimizes energy loss during the switching process.
    - **Linear Throttle Response:** Provides a more direct relationship between the input signal and the motor speed.

Features of Simonk Firmware

- High refresh rate
- Damped light (for regenerative braking in some versions)
- Improved motor response
- Optimized for multirotors

Applications

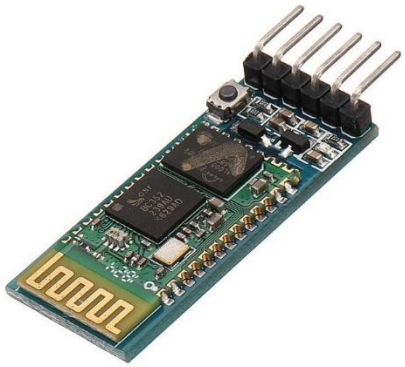
- Multirotors (drones)
- RC airplanes
- RC helicopters
- Robotics

Use in Our Project

In our wall-climbing window-cleaning robot, the Simonk 30A ESC is used to control the BLDC motor that generates suction:

- The ARDUINO Uno sends a control signal to the Simonk 30A ESC.
- The ESC regulates the speed of the 2450KV BLDC motor.

5.7 HC-05 Bluetooth Module



The HC-05 is a popular and inexpensive Bluetooth module that enables wireless serial communication between devices, such as microcontrollers (like ARDUINO) and smartphones or computers.

Introduction

The HC-05 module allows your ARDUINO or other microcontroller to communicate wirelessly using the Bluetooth protocol. It's commonly used for remote control, data transfer, and other applications where a short-range wireless connection is needed. The HC-05 can operate in both master and slave modes.

Specifications

Feature	Specification	Details
Bluetooth Version	2.0 + EDR	Enhanced Data Rate, providing up to 3 Mbps data transfer.
Operating Voltage	3.3V - 5V	Typically operates at 3.3V, but many modules have an on-board regulator to accept 5V.
Operating Current	< 50mA	Low power consumption.
Frequency Band	2.4GHz ISM	Industrial, Scientific, and Medical frequency band.
Modulation	GFSK	Gaussian Frequency Shift Keying.
Output Power	-23dBm to +4dBm	
Sensitivity	-80dBm	
Range	~10 meters	Varies depending on environmental conditions and other factors.
Interface	Serial (UART)	Uses serial communication to exchange data with the microcontroller.
Baud Rate	Configurable	Can be set to various speeds (e.g., 9600 bps, 38400 bps, etc.).
Modes	Master or Slave	Can initiate a connection (master) or wait for a connection (slave).

Pin Details	VCC	Power supply (3.3V or 5V)
	GND	Ground
	TXD	Transmit Data (to microcontroller RX)
	RXD	Receive Data (from microcontroller TX)
	KEY	PIO8 (optional)
	STATE	PIO9 (optional)

Working Principle

The HC-05 module essentially acts as a wireless serial port. It sends and receives data over the Bluetooth connection, and this data is then transmitted and received by the microcontroller via its UART interface.

1. **Serial Communication:** The ARDUINO communicates with the HC-05 using its serial transmit (TX) and receive (RX) pins.
2. **Bluetooth Connection:**
    - When powered on, the HC-05 module can be configured to either wait for a connection from another Bluetooth device (slave mode) or initiate a connection to another device (master mode).
    - In the project, the HC-05 is likely configured as a slave, waiting for a connection from your smartphone.
3. **Data Transfer:**
    - Once a Bluetooth connection is established, any data sent from the smartphone (or other Bluetooth device) is received by the HC-05 and transmitted to the ARDUINO's RX pin via serial communication.
    - Similarly, any data sent by the ARDUINO's TX pin is transmitted by the HC-05 over the Bluetooth connection to the connected device.
4. **Command Mode:** The HC-05 can be placed in a command mode where we can configure its settings (e.g., baud rate, name, pairing password) using a set of AT commands.

Key Features

- Wireless serial communication
- Easy to use with microcontrollers
- Low cost
- Configurable settings

Applications

- Wireless control of robots
- Wireless data transfer
- Remote control of devices
- Home automation
- Wireless sensor networks

Use in Our Project

In our wall-climbing window-cleaning robot, the HC-05 Bluetooth module is used to enable wireless control of the robot's movement and steering:

- The HC-05 module is connected to the ARDUINO Uno's serial pins (TX and RX).
- A smartphone with a Bluetooth control app is used to send commands to the robot.
- The smartphone sends commands (e.g., 'F' for forward, 'B' for backward, 'L' for left, 'R' for right) over Bluetooth.
- The HC-05 receives these commands and transmits them to the ARDUINO via serial communication.
- The ARDUINO then interprets these commands and controls the L298N motor driver (for movement) and the servo motor (for steering).

5.8 12V 3S 30C LiPo Battery



A 12V 3S 30C LiPo battery is a type of rechargeable lithium polymer battery commonly used in applications requiring high power and energy density, such as RC vehicles, drones, and robotics.

Introduction

LiPo (Lithium Polymer) batteries are popular due to their high energy density, lightweight nature, and ability to deliver high currents. The "3S" configuration, "12V" nominal voltage, and "30C" discharge rating are important characteristics.

Specifications

Feature	Specification	Details
Battery Type	Lithium Polymer (LiPo)	
Configuration	3S	3 cells connected in series.
Nominal Voltage	11.1V	The nominal voltage of a single LiPo cell is 3.7V. 3 cells in series: 3 * 3.7V = 11.1V. Often referred to as "12V" in common usage.
Maximum Voltag	12.6V	Fully charged voltage (4.2V per cell).
Minimum Voltag	9.0V	Safe discharge cutoff voltage (3.0V per cell). Discharging below this can damage the battery.
Capacity	1300 mAh	1300 milliamp-hours. This indicates how much charge the battery can store. It can supply 1.3 Amps for 1 hour, or fractions thereof.
Discharge Rate	30C	"C" rating indicates the rate at which the battery can be safely discharged.
Maximum Continuous Discharge Curren	39A	30C * 1.3Ah = 39A. The battery can continuously supply 39 Amps.
Connector Type	Varies (e.g., XT60)	The connector used to connect the battery to the PDB or other devices.

Understanding the Specifications

- **3S Configuration:**
  - "3S" means the battery is made up of 3 individual LiPo cells connected in series.
  - Connecting cells in series increases the voltage. Each LiPo cell has a nominal voltage of 3.7V, so a 3S battery has a nominal voltage of 11.1V.
- **12V Nominal Voltage:**
  - While the precise voltage is 11.1V, LiPo batteries are often referred to by their nominal voltage. A 3S LiPo is commonly called a "12V" battery.
- **1300 mAh Capacity:**
  - "mAh" stands for milliamp-hours.
  - A 1300 mAh battery can theoretically provide 1300 milliamps (1.3 Amps) for one hour.
  - Or, it could provide 2600 milliamps (2.6 Amps) for 30 minutes, and so on.
  - It's a measure of how much energy the battery can store.
- **30C Discharge Rate:**
  - The "C" rating indicates how quickly the battery can be safely discharged.
  - A 30C rating means the battery can be discharged at a rate that is 30 times its capacity.
  - In this case: 30C * 1.3 Ah = 39 Amps. The battery can provide a continuous current of 39 Amps.
  - Exceeding the C rating can damage the battery, cause it to overheat, or even cause a fire.

LiPo Battery Safety

LiPo batteries require careful handling due to their potential for fire or explosion if mishandled. It is crucial to:

- Use a LiPo-specific charger.
- Never overcharge or over-discharge the battery.
- Store the battery in a fireproof bag.
- Avoid physical damage to the battery.
- Monitor the battery during charging.

Applications

- RC vehicles (cars, planes, boats)
- Drones and multirotors
- Robotics
- Portable power supplies

Use in Our Project

In our wall-climbing window-cleaning robot, the 12V 3S 30C 1300 mAh LiPo battery is used as the main power source for the entire robot:

- It provides the necessary power to drive all the robot's components:
  - The N20 motors (via the L298N motor driver)
  - The ARDUINO Uno
  - The Simonk 30A ESC (for the BLDC motor)
  - The HC-05 Bluetooth module
- The high discharge rate (30C) allows the battery to supply the current required by the motors, especially the BLDC motor, which draws a significant amount of power for suction.
- The XT60 connector on the battery is used to connect it to the PDB, which then distributes the power to the various components.

5.9 Propeller and Suction Details for Wall-Climbing Robot



A propeller is a crucial component in our wall-climbing window-cleaning robot. Its rotation generates the suction force that enables the robot to adhere to the vertical window surface.

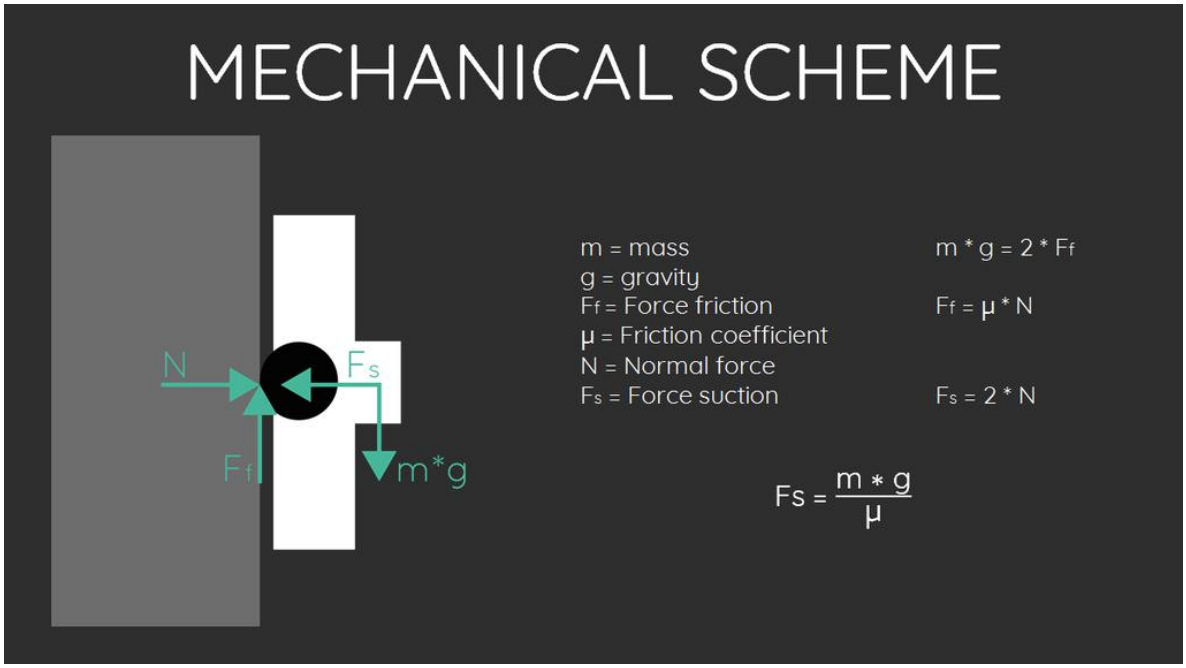
Propeller Specifications (1045)

Feature	Specification	Details
Diameter	10 inch	The diameter of the circle swept by the propeller blades.
Pitch	4.5 inch	The theoretical distance the propeller would advance in one revolution.
Designation	1045	A common way to refer to propellers, where the first two digits are the diameter and last two are the pitch
Number of Blades	2	The number of blades on the propeller.
Material	Plastic	Lightweight and easy to manufacture.
Rotation Direction	Clockwise (or Counter-Clockwise)	The direction in which the propeller rotates.

Thrust and Suction: How the Propeller Creates Adhesion

1. **Airflow and Thrust:** When the BLDC motor spins the propeller, the propeller blades accelerate air. This acceleration of air creates a force called *thrust.*
2. **Direction of Thrust:** The propeller is oriented so that it pushes air *away* from the robot's base (the part in contact with the window).
3. **Pressure Differential:**
    - As the propeller pushes air away, it creates a region of *lower air pressure* between the robot's base and the window surface.
    - Simultaneously, the air pressure on the *outside* of the robot (the side away from the window) remains at normal atmospheric pressure.
4. **Suction Force:** This difference in air pressure results in a net force pushing the robot *towards* the window. We call this force the *suction force*. It's analogous to how a vacuum cleaner works. The atmosphere is pushing the robot against the window.
5. **Equilibrium:** The robot adheres to the window when the suction force generated by the propeller is equal to or greater than the forces trying to pull it away from the window (primarily gravity and any friction from movement).

Adhesion Force Calculations



Adhesion Force Calculation

To ensure the robot adheres securely to vertical surfaces, we need to calculate the **minimum suction force (Fs)** required to counteract the gravitational force and any slipping due to surface friction.

We'll use the friction-based equilibrium formula:

$$F_s = \frac{m \cdot g}{\mu}$$

Where:

- F_s = Required suction force (N)
- m = Mass of the robot = **1.2 kg** *(measured)*
- g = Acceleration due to gravity = **9.81 m/s²**
- μ = Coefficient of friction = **0.6** *(rubber on glass/metal)*

Step-by-Step Calculation:

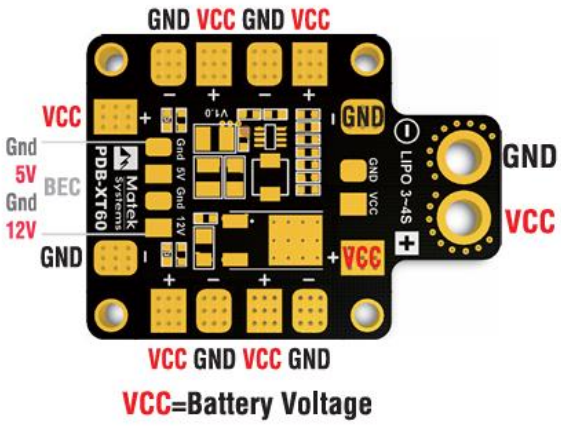
$$F_s = \frac{1.2 \cdot 9.81}{0.6}$$

$$F_s = \frac{11.772}{0.6}$$

$$F_s = 19.62\,\text{N}$$

$$\boxed{F_s \approx 2.0\,\text{kgf (kilogram-force)}}$$

5.10 PDB (Power Distribution Board) XT60





A PDB (Power Distribution Board) is a circuit board that simplifies the distribution of power from a battery to multiple components in a multirotor or other electronic project. The XT60 connector is a common type of connector used for high-current connections, often found on PDBs.

Specifications

Since "PDB XT60" refers to a combination of components (a board and a connector), the specifications describe the general characteristics of a typical PDB with an XT60 connector, and the connector itself.

Feature	Specification	Details
PDB Function	Power Distribution	Distributes power from a single battery to multiple devices.
Input Connector	XT60 Connector	A high-current connector.
Output Connectors	Solder Pads or Connectors	Provides connections for ESCs, cameras, etc. The number and type vary.
Voltage Range	Varies (up to battery voltage)	The PDB handles the voltage of the connected battery (e.g., 2S-6S LiPo).
Current Rating	Varies (e.g., 100A or more)	The PDB is designed to handle high currents.
BEC (Optional)	Some PDBs have a BEC	A Battery Eliminator Circuit steps down the battery voltage to a lower voltage

		(e.g., 5V or 12V) for powering other components.
Filtering (Optional)	Some PDBs have filtering	Components to reduce electrical noise.
Fuses (Optional)	Some PDBs have fuses	To protect circuits.
XT60 Connector	Connector Type	XT60
	Current Rating	60A continuous
	Voltage Rating	Up to battery voltage
	Polarity	Polarized (prevents reverse connection)

Working Principle

The PDB itself is a relatively simple device. It consists of conductive traces (thick copper lines) that route power from the battery input to multiple output points.

1. **Battery Connection:** The battery is connected to the PDB via the XT60 connector.
    - The XT60 connector ensures a secure and low-resistance connection, capable of handling high currents without overheating.
2. **Power Distribution:** The PDB's conductive traces distribute the battery voltage to the various output points.
    - These output points are typically solder pads or connectors where you can connect ESCs, cameras, and other devices.
3. **Voltage Regulation (Sometimes):** Some PDBs include a BEC (Battery Eliminator Circuit).
    - A BEC is a voltage regulator that takes the higher voltage from the main battery and steps it down to a lower voltage (e.g., 5V or 12V).
    - This lower voltage can then be used to power the receiver, flight controller, and other sensitive electronic components.
4. **Filtering and Protection (Sometimes):** Some PDBs may also include:
    - Capacitors to filter out electrical noise, which can interfere with the operation of sensitive electronics.

- Fuses or other protection circuits to prevent damage from overcurrent or short circuits.

Applications

- Multirotors (drones)
- RC airplanes and helicopters
- Robotics projects with multiple motors and devices
- Any project requiring organized power distribution

Use in Our Project

In our wall-climbing window-cleaning robot, the PDB XT60 is used to:

- Connect the 12V LiPo battery to the robot's electrical system.
- Distribute power from the battery to the various components:
  - The L298N motor driver (for the N20 motors)
  - The ARDUINO Uno
  - The Simonk 30A ESC (for the BLDC motor)
  - The HC-05 Bluetooth module
- The XT60 connector provides a reliable, high-current connection between the battery and the PDB.

5.11 N20 Motor



N20 motors are small, geared DC motors commonly used in robotics, model making, and other applications where compact size and moderate torque are required.

Introduction

N20 motors are popular due to their small form factor, relatively high torque output (thanks to the gearbox), and availability in various configurations. The gearbox is a crucial part of the motor assembly.

Specifications

Feature	Specification (General)	Details
Motor Type	Brushed DC Motor	
Gearbox Type	Planetary Gearbox	Reduces the output speed and increases torque.
Operating Voltage	1.5V - 12V	
No-Load Speed	Varies (e.g., 30RPM to 1000RPM)	Speed of the motor shaft when no load is applied.
No-Load Current	Varies (e.g., 10mA to 100mA)	Current drawn with no load.
Stall Torque	Varies (e.g., 0.1 kg-cm to 5 kg-cm)	Torque when the motor shaft is completely stopped. This is the maximum torque the motor can exert.

Stall Current	Varies (e.g., 0.5A to 2A)	Current drawn when the motor is stalled.
Output Shaft Diameter	3mm	
Output Shaft Length	Varies	
Size	12mm x 10mm x L (motor) + gearbox length	Compact dimensions.
Weight	Varies	
Wiring	Two wires	Connect to motor driver. Reversing polarity changes direction.

Working Principle

An N20 motor combines a small DC motor with a gearbox.

1. **DC Motor:** The core of the N20 is a small brushed DC motor.
    - When voltage is applied, the motor's internal rotor rotates due to the interaction of magnetic fields.
    - Brushes and a commutator transfer electrical current to the motor windings.
2. **Planetary Gearbox:** The DC motor's high-speed, low-torque output is fed into a planetary gearbox.
    - Planetary gears consist of a central "sun" gear, several "planet" gears that rotate around the sun gear, and an outer "ring" gear.
    - This arrangement reduces the output speed and increases the output torque.
    - The gear ratio determines how much the speed is reduced and the torque is increased. A higher gear ratio means lower speed and higher torque.
3. **Output:** The gearbox's output shaft rotates at a lower speed but with significantly higher torque than the DC motor itself.

Advantages of N20 Motors

- Small size
- Relatively high torque for their size
- Low cost
- Available in many speed/torque configurations

Applications

- Robotics: Driving wheels, arms, and other actuators.
- Model making: Controlling movement in model vehicles and toys.
- Consumer electronics: Actuating small mechanisms.
- Door locks
- Small pumps

Use in Our Project

In our wall-climbing window-cleaning robot, the N20 motors are used to provide the driving force for movement along the window:

- Two N20 motors are used to drive the rear wheels of the robot.
- The L298N motor driver controls the speed and direction of these motors.
- The relatively high torque of the N20 motors (due to their gearboxes) allows the robot to move on the vertical window surface, overcoming friction and the weight of the robot.The ARDUINO controls the L298N, which in turn controls the N20 motors, to move the robot forward, backward, and stop.

5.12 Mechanical Structure Frame Design



- The frame of the robot is constructed using **Forex sheet**, which is lightweight, rigid, and easy to cut, making it ideal for custom robotic builds.
- The total **fan blade length is 10 inches**, and the frame is designed to match this dimension for proper suction coverage and stability.
- **Front 5 inches** of the frame: Allocated for mounting essential components like the Arduino, sensors, and Bluetooth module.
- **Rear 2 inches**: Used to place the motor driver and associated circuitry.
- **2-inch extensions on both the left and right sides**: Provided to ensure balanced structure and even weight distribution.
- The **main frame is cut in a plus ("+") shape**, with a **1-inch central cut-out** to securely fix the BLDC motor used for suction.
- This layout allows proper airflow and effective suction across the 10-inch fan span, essential for wall-climbing functionality.
- To minimize vibrations and bouncing, a **supporting sub-frame** was added at the rear side.
- This was done by pasting an extra bit of Forex sheet to the back of the main frame, reinforcing the structure.
- The supporting section improves mechanical rigidity and stabilizes the robot during vertical movement.

6. SOFTWARE

6.1 Software Details

The software component of the ARDUINO-based wall-climbing glass cleaning robot is crucial for its operation, managing motor control, sensor data, user interaction through Bluetooth, and safety protocols. The ARDUINO Uno R3 microcontroller serves as the robot's "brain," and it is programmed using the ARDUINO IDE (Integrated Development Environment). The software dictates the robot's actions based on information from its sensors and commands from the user.

6.2 Code

```
#include <Servo.h> // Required for ESC control

#define TRIG_PIN 4
#define ECHO_PIN 5

#define GREEN_LED A0
#define RED_LED A1
#define BUZZER_PIN 11 // Moved from A2 to D11 (PWM-capable digital pin)

// Rear motor pins
#define IN1 6
#define IN2 7
#define IN3 8
#define IN4 9

Servo bldc; // BLDC ESC control
float duration_us, distance_cm;
String command = "";

int currentPWM = 1000; // Track current speed for smooth ramping

void setup() {
  Serial.begin(9600);

  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  pinMode(GREEN_LED, OUTPUT);
  pinMode(RED_LED, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);

  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);
```

```
  bldc.attach(3); // ESC PWM signal pin
  bldc.writeMicroseconds(1000); // Arm ESC
  delay(3000);

  Serial.println("Ready with 1 Motor Driver ✅ + BLDC control");
}

void forward() {
  digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
  digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
}

void backward() {
  digitalWrite(IN1, LOW); digitalWrite(IN2, HIGH);
  digitalWrite(IN3, LOW); digitalWrite(IN4, HIGH);
}

void stopMotors() {
  digitalWrite(IN1, LOW); digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW); digitalWrite(IN4, LOW);
}

void turnLeft() {
  digitalWrite(IN1, LOW); digitalWrite(IN2, HIGH);
  digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
}

void turnRight() {
  digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW); digitalWrite(IN4, HIGH);
}

void playBeep(int delayGap) {
  tone(BUZZER_PIN, 1000); delay(50);
  noTone(BUZZER_PIN); delay(delayGap);
}

void rampToSpeed(int targetPWM) {
  Serial.print("Ramping to ");
  Serial.print(targetPWM);
  Serial.println(" µs");

  if (targetPWM > currentPWM) {
    for (int i = currentPWM; i <= targetPWM; i++) {
      bldc.writeMicroseconds(i);
      delay(5); // Adjust smoothness
    }
  } else if (targetPWM < currentPWM) {
    for (int i = currentPWM; i >= targetPWM; i--) {
      bldc.writeMicroseconds(i);
```

```
      delay(5); // Adjust smoothness
    }
  }

  currentPWM = targetPWM;
}

void loop() {
  // Distance measure
  digitalWrite(TRIG_PIN, LOW); delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH); delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);
  duration_us = pulseIn(ECHO_PIN, HIGH);
  distance_cm = 0.017 * duration_us;

  // LED logic
  if (distance_cm > 15) {
    digitalWrite(GREEN_LED, HIGH);
    analogWrite(RED_LED, 0);
  } else {
    digitalWrite(GREEN_LED, LOW);
    int redIntensity = map(distance_cm, 15, 0, 0, 255);
    analogWrite(RED_LED, constrain(redIntensity, 0, 255));
  }

  // Buzzer for obstacle detection
  if (distance_cm < 15) {
    int delayGap = 0;
    if (distance_cm > 10) delayGap = 300;
    else if (distance_cm > 7) delayGap = 150;
    else if (distance_cm > 4) delayGap = 80;
    else delayGap = 30;

    playBeep(delayGap);
  }

  // Bluetooth command
  while (Serial.available()) {
    char ch = Serial.read();
    ch = tolower(ch);

    if (ch == 'f') {
      forward(); Serial.println("Forward");
    } else if (ch == 'b') {
      backward(); Serial.println("Backward");
    } else if (ch == 's') {
      stopMotors(); Serial.println("Stop");
    } else if (ch == 'l') {
      turnLeft(); Serial.println("Turning Left (Tank Turn)");
    } else if (ch == 'r') {
      turnRight(); Serial.println("Turning Right (Tank Turn)");
    } else if (ch == 'y') {
```

```
      Serial.println("Horn Honk!");
      tone(BUZZER_PIN, 1000); delay(100);
      noTone(BUZZER_PIN); delay(100);
      tone(BUZZER_PIN, 1000); delay(150);
      noTone(BUZZER_PIN);
    }

    // BLDC smooth speed control
    else if (ch == '1') {
      rampToSpeed(1300); Serial.println("BLDC Speed: LOW");
    } else if (ch == '2') {
      rampToSpeed(1350); Serial.println("BLDC Speed: MEDIUM");
    } else if (ch == '3') {
      rampToSpeed(1400); Serial.println("BLDC Speed: HIGH");
    } else if (ch == '4') {
      rampToSpeed(1500); Serial.println("BLDC Speed: FULL");
    } else if (ch == 'z') {
      rampToSpeed(900); Serial.println("BLDC STOPPED");
    }
  }
}
```

6.3 Code Description

The software is structured using libraries, variables, and functions, each with a specific role.

- **Libraries:**
  - <Servo.h>: This library is used to control the speed of the BLDC motor, which provides the suction force. While it's a "servo" library, it's suitable because the BLDC motor's controller (ESC) is controlled using similar signals.
- **Pin Definitions:**
  - Specific digital pins on the ARDUINO are designated for the ultrasonic sensor (to measure distance), the green and red LEDs (for visual feedback), the buzzer (for audible alerts), and the motor driver (to control movement).
- **Variables:**
  - bldc: This variable is used to send commands to the BLDC motor's speed controller.
  - duration_us, distance_cm: These variables store the data from the ultrasonic sensor, representing the time it takes for the sound wave to return and the calculated distance to an obstacle.
  - command: This variable stores commands received wirelessly via Bluetooth from the user's control device.
  - currentPWM: This variable keeps track of the BLDC motor's current speed setting, allowing for smooth changes in speed.
- **Functions:**
  - setup(): This function runs once when the robot is turned on. It initializes communication, sets up the pins for each component, and prepares the BLDC motor's controller.
  - forward(), backward(), stopMotors(), turnLeft(), turnRight(): These functions control the direction of the robot's movement by sending signals to the motor driver, which in turn controls the N20 motors.
  - playBeep(): This function creates a beeping sound using the buzzer, used to alert the user to events or warnings.
  - rampToSpeed(): This function smoothly increases or decreases the BLDC motor's speed, preventing sudden changes that could affect adhesion.
  - loop(): This function runs continuously while the robot is operating. It handles sensor readings, Bluetooth commands, and motor control.

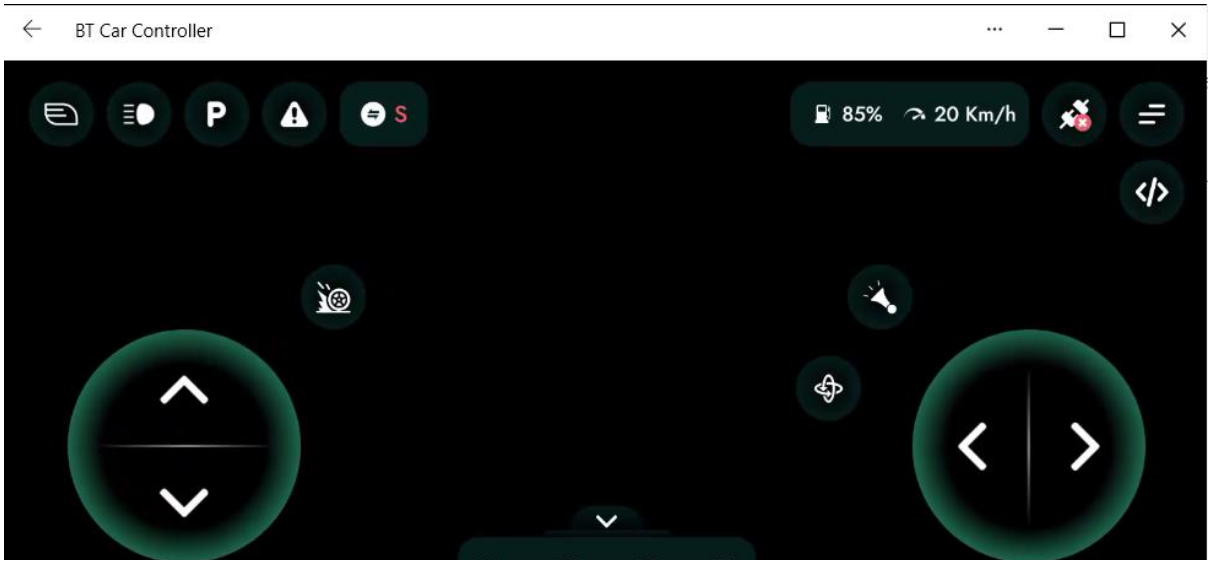
6.4 Code Functionality and Operation

The software operates in a sequence of steps:

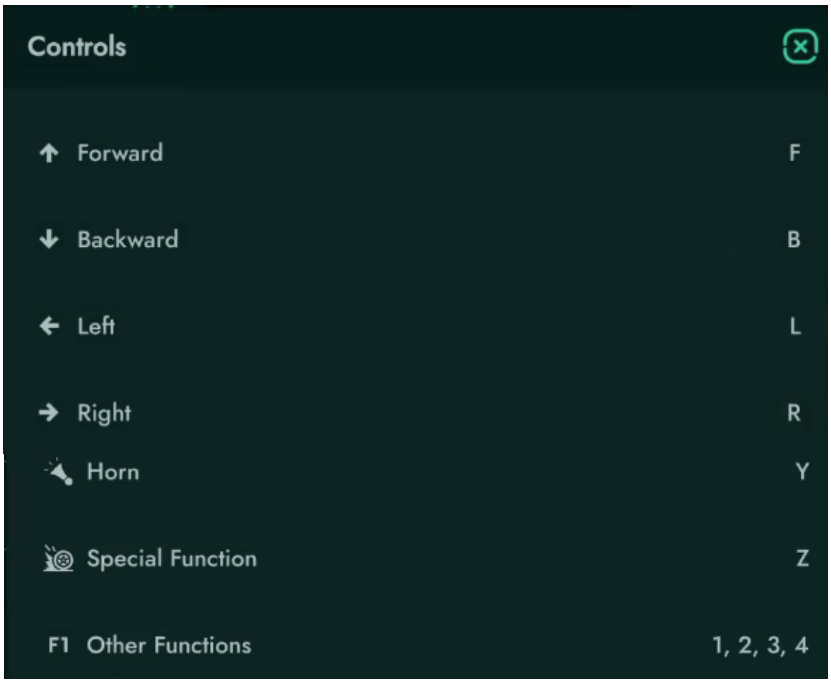
1. **Initialization (**setup()**):** When the robot is powered on, the setup() function prepares all the components. This includes setting up communication with the Bluetooth module and configuring the input and output pins for the sensors, motors, and indicators.
2. **Continuous Operation (**loop()**):** The loop() function then takes over, running repeatedly to control the robot:
    - **Distance Measurement:** The ultrasonic sensor measures the distance to any nearby obstacles, allowing the robot to "see" its environment.
    - **Feedback and Alerts:** The LEDs and buzzer provide feedback to the user. For example, the red LED might brighten as the robot approaches an obstacle, and the buzzer might sound a warning.
    - **Bluetooth Control:** The software constantly checks for commands sent from the user via Bluetooth. These commands can control the robot's movement (forward, backward, turning) and the BLDC motor's speed (suction strength).
3. **Motor Control:** The robot's movement is achieved by sending signals to the motor driver, which powers the N20 motors. The BLDC motor's speed, and therefore the suction force that keeps the robot attached to the glass, is precisely controlled.
4. **Obstacle Detection:** The ultrasonic sensor enables the robot to detect obstacles, allowing for basic avoidance maneuvers or stopping to prevent collisions.
5. **Wireless Control:** Bluetooth communication provides the user with remote control over the robot, increasing its flexibility and ease of use.

In summary, the software acts as the intelligent control system of the robot, enabling it to move, adhere to surfaces, respond to its environment, and be operated wirelessly by a user.

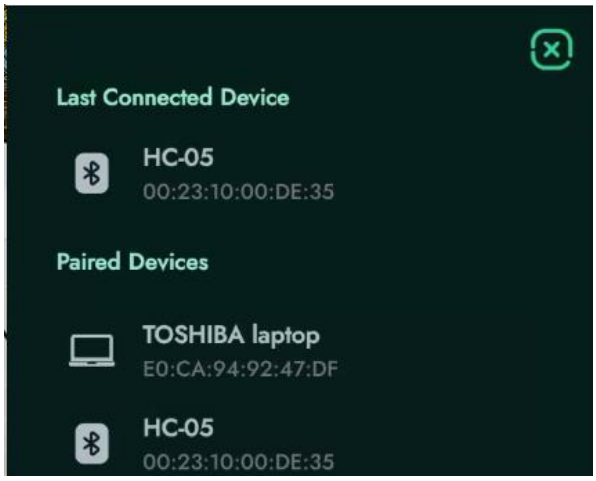
6.5 Bluetooth Control Application



1 App Interface



3 App Controls and function characters



2. App device connection list

A custom-developed Bluetooth control application enables users to wirelessly operate the wall-climbing robot. This application provides a user-friendly interface for controlling the robot's movement and suction force, enhancing its practicality and operational flexibility.

The application's graphical user interface (GUI) is designed for intuitive control and clear feedback, featuring several key control elements:

- **Movement Controls:** The primary interface incorporates both joystick-style controls and a discrete button overlay ('Forward', 'Backward', 'Left', 'Right') for precise robot maneuvering (Figs.1,2). These controls allow the user to navigate the robot across the cleaning surface with accuracy.
- **BLDC Motor Control:** Stepped control of the BLDC motor speed, and consequently the robot's suction force, is achieved through buttons labeled 'F1' to 'F4'. This functionality enables the user to optimize the suction for varying surface conditions and cleaning requirements. A 'Special Function' button is included to immediately stop the BLDC motor.
- **Auditory Feedback:** To provide user alerts, a 'Horn' button activates the robot's buzzer, generating an audible signal.
- **Bluetooth Connectivity:** The application simplifies the connection process with an integrated device list. This list displays both recently connected and paired Bluetooth devices, facilitating easy selection and connection to the robot's HC-05 module (Fig. 3).
- **Status Indicators:** While the main control screen includes status indicators, their specific functionality is not directly relevant to the robot's control operations.

Key Improvements in this Combined Version:

- **Strong Opening:** Starts with a clear and concise statement of the app's purpose.
- **Combined Strengths:** Merges the functional descriptions with the user interface details.
- **Improved Flow:** Presents the information in a logical and easy-to-follow sequence.
- **Emphasis on Usability:** Highlights the user-friendly design of the application.
- **Complete Information:** Includes all relevant details about movement control, suction control, feedback, and connectivity.

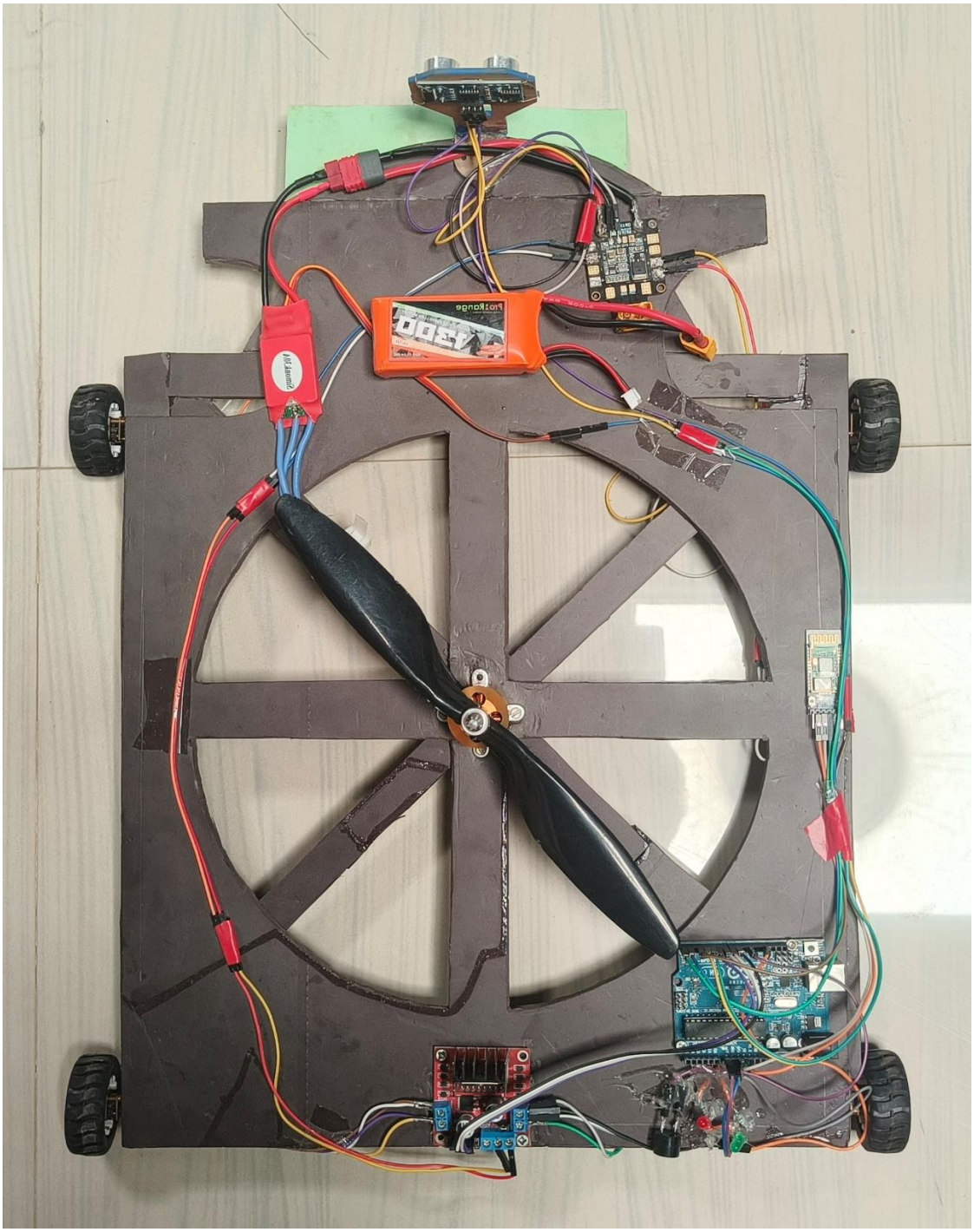
6.6 Bluetooth App Button Commands

Button/Control	Command Sent	Robot Action	Description
Forward	'f'	Robot moves forward	Drives the robot in a forward direction.
Backward	'b'	Robot moves backward	Drives the robot in a backward direction.
Left	'l'	Robot turns left	Turns the robot to the left
Right	'r'	Robot turns right	Turns the robot to the right
Stop	's'	Robot stops moving	Stops all N20 motor movement.
Horn	'y'	Buzzer sounds	Activates the buzzer to produce a "horn" sound.
Special Function	'z'	BLDC motor stopped	Stops the BLDC motor.
F1	'1'	BLDC motor speed low	Sets the BLDC motor to low speed.
F2	'2'	BLDC motor speed medium	Sets the BLDC motor to medium speed.
F3	'3'	BLDC motor speed high	Sets the BLDC motor to high speed.
F4	'4'	BLDC motor speed full	Sets the BLDC motor to full speed.

Explanation of the Table:

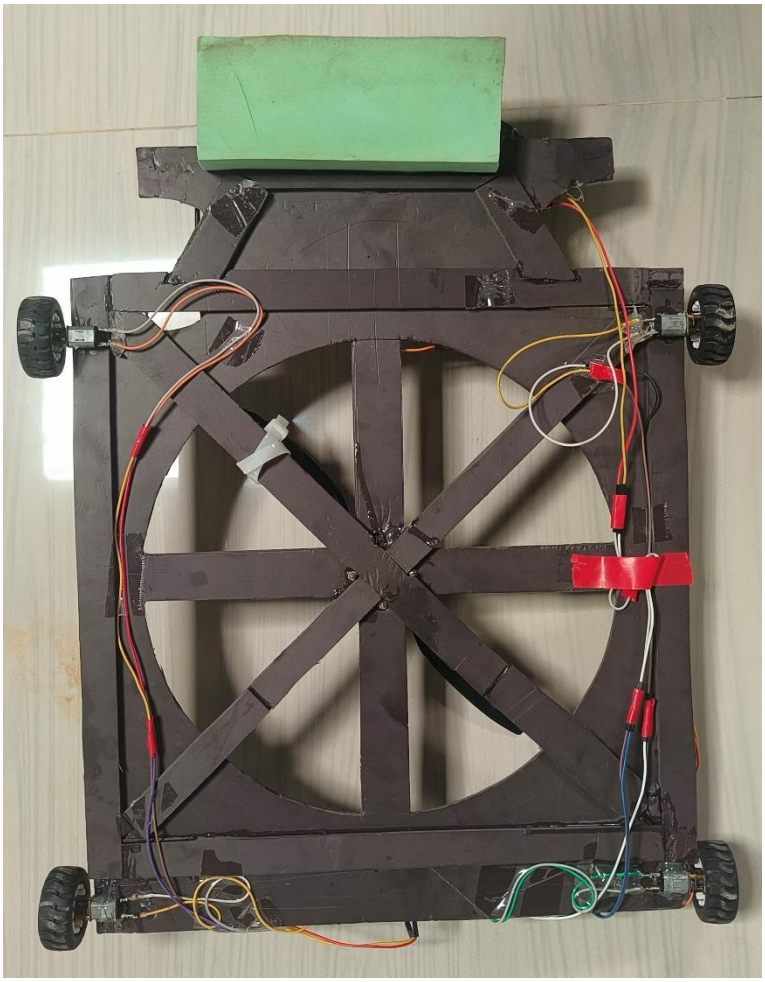
- **Button/Control:** This column refers to the button or control element in the Bluetooth app's user interface.
- **Command Sent:** This column indicates the character or string that the app sends to the ARDUINO via Bluetooth when the corresponding button is pressed.
- **Robot Action:** This column describes the action that the robot performs in response to the received command.
- **Description**: More detailed explanation of the action.

7. Project Photos

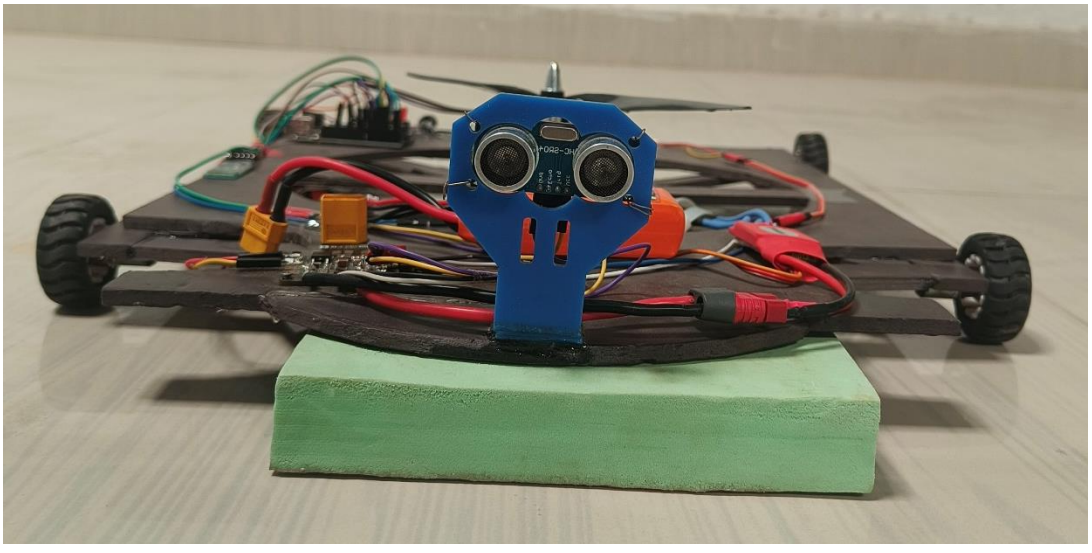


1. TOP VIEW

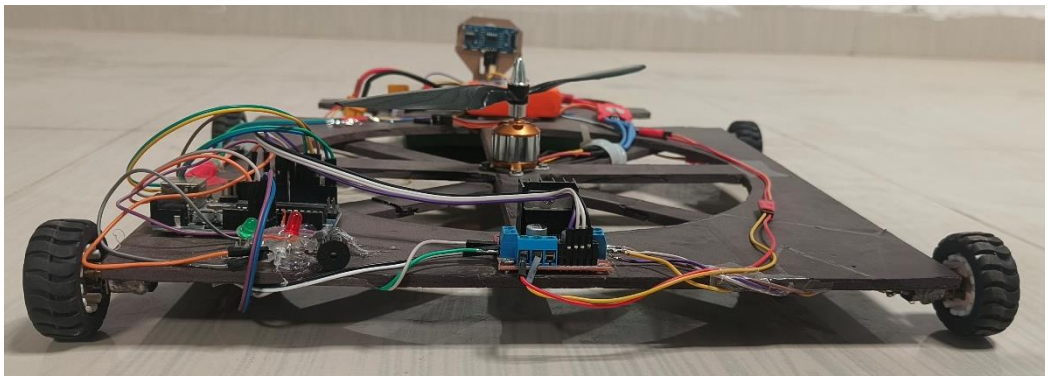
2. BOTTOM VIEW



3. FRONT VIEW



4. BACK VIEW



6. RIGHT SIDE VIEW



5. LEFT SIDE VIEW



8. Conclusion

The Arduino-based vertically wall-climbing glass-cleaning robot developed in this project successfully addresses the need for a safer, more efficient method of cleaning vertical glass surfaces, especially in high-rise structures. By integrating suction-based adhesion using a 2450KV BLDC motor, movement control with N20 gear motors, and obstacle detection via ultrasonic sensors, the robot achieves stable and responsive performance on vertical planes.

The system is wirelessly controlled through Bluetooth, allowing remote operation and user-friendly interaction. The project also includes visual and auditory feedback using LEDs and a buzzer for enhanced operational safety. All components were carefully selected, integrated, and tested to ensure functionality, power efficiency, and structural balance.

Throughout development, the team overcame significant challenges such as BLDC motor overheating, communication issues with the ESC, and mechanical design limitations. These were resolved through innovative solutions including servo tester-based ESC control and two-wheel optimization.

This prototype sets a strong foundation for future improvements like automated path planning, rotating sponge cleaning mechanisms, and battery level monitoring. With further refinement, the robot has the potential to evolve into a fully autonomous cleaning solution for commercial and residential applications.

9. Future Scope

The wall-climbing glass-cleaning robot, while effective in its current form, presents multiple opportunities for enhancement and scalability:

- **Autonomous Navigation**: Future versions can integrate real-time path planning algorithms using advanced sensors (e.g., gyroscope, IMU, or camera vision) to enable fully autonomous movement without manual input.
- **Rotating Cleaning Mechanism**: Incorporating a rotating sponge or brush system, driven by an additional motor, would improve cleaning efficiency and surface coverage, especially for stubborn stains or large panels.
- **Battery Health Monitoring**: Implementing voltage and current sensing circuits can help monitor battery health and prevent sudden shutdowns, improving safety and performance reliability.
- **Mobile App Development**: A custom Android/iOS application with graphical controls, battery status display, and sensor feedback could provide more intuitive control and system monitoring.
- **Enhanced Adhesion**: Using dual-propeller suction or vacuum pump systems can generate stronger and more stable adhesion, allowing the robot to climb larger surfaces and even operate outdoors under wind disturbances.
- **AI Integration**: Leveraging machine learning for obstacle classification or adaptive path correction can make the system smarter and more efficient over time.
- **Material and Weight Optimization**: Redesigning the chassis with lighter, stronger materials like carbon fiber or aluminum alloy can improve mobility, reduce power consumption, and increase overall reliability.

These future enhancements would transform the robot from a semi-autonomous prototype into a robust, commercial-grade solution for industrial and residential window cleaning.

10. Limitations of the Project

While the project achieved its core objectives, certain limitations were identified during development and testing:

- The suction force generated by the 2450KV BLDC motor was slightly below the ideal threshold, limiting adhesion reliability on dusty or uneven surfaces.
- Continuous operation caused the motor and ESC to heat up, requiring cooling breaks to prevent component stress or failure.
- The robot relies on manual Bluetooth control and lacks full autonomous navigation, making it dependent on user input for movement and cleaning coverage.
- Obstacle detection is unidirectional and limited in range, affecting its ability to navigate around large or complex obstructions.
- The battery life restricts operation time, especially when running both suction and drive motors simultaneously under load.

These limitations highlight areas for future improvement to enhance the robot's real-world viability and operational safety.

11. Applications

The wall-climbing glass-cleaning robot has wide-ranging applications across both residential and industrial environments, including:

- High-rise building window cleaning where human access is dangerous or impractical.
- Commercial buildings with large glass facades, such as malls, hotels, and corporate offices.
- Maintenance in laboratories, hospitals, and cleanrooms where consistent hygiene is essential.
- Cleaning of large interior glass walls and decorative panels in luxury residential settings.
- Surveillance or inspection use in elevated or hazardous environments, after modification.

This robot demonstrates how automation can reduce human risk while improving cleaning consistency and efficiency.

12. Summary

This project presents the successful design and implementation of an Arduino-based vertically wall-climbing robot for automated glass cleaning.

By combining BLDC suction-based adhesion, gear motor-driven mobility, ultrasonic obstacle detection, and Bluetooth remote control, the prototype meets its intended functionality while addressing real-world safety and efficiency concerns.

The system was built using off-the-shelf components and custom logic, offering a cost-effective and scalable solution for modern maintenance challenges.

Despite certain limitations, the project lays a strong foundation for advanced robotic systems capable of operating in vertical and high-risk environments.

13. Cost Analysis

Here's a cost analysis for the ARDUINO-based window cleaning robot:

S.No	Component Name	Cost (Rs)
1	ARDUINO Uno R3	420
2	L298N Motor Driver	180
3	Ultrasonic Sensor	80
4	Buzzer	40
5	Bluetooth Module (HC-05)	250
6	Chassis/Frame	250
7	Battery	1300
8	Wiring and Connectors	300
9	Cleaning Mechanism	60
10	PD board	300
11	N20 motor and wheel	660
12	Propeller	180
13	BLDC motor and ESC	930
14	Cleaning Sponge	50
15	Glue cutter	100
	Total	5,150

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- Here you go — ready to **copy-paste directly** into your report's **References** section at the end:
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- L298N Motor Driver Datasheet. Available at: https://www.st.com/resource/en/datasheet/l298.pdf

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